

Anutej Sachin Kardele

anutej.kardele@gmail.com • +1 (716) 432-4660 • Buffalo, NY • anutej.us • LinkedIn • GitHub

EXPERIENCE

Software Engineering Practicum, HeinOnline, Buffalo, NY

Jan 2026 – May 2026

- Engineered a Python-powered OCR and AI extraction pipeline leveraging Qwen language models, performing ablation studies to optimize multi-document processing; achieved a 90.58% extraction accuracy across 28,167 journal pages.
- Built services to cross-reference extracted PDF metadata against Crossref APIs, mapping verified author records and publication metadata (ISSN, date, volume) into PostgreSQL databases to automate Flipper production ingestion.

Software Engineer, Simbott Technologies, Mumbai, India

Mar 2024 - Jul 2025

- Architected a scalable, component-based C# architecture for enterprise applications, using Git to reduce duplicated code by 80% and enable custom module implementation in under 5 minutes within secure environments.
- Co-engineered 3 custom peripherals in collaboration with the electronics team, authoring high-performance firmware to establish low-latency serial communication pipelines between edge hardware and the central simulation engine.
- Engineered real-time telemetry processing pipelines for custom physical controllers, optimizing sensor data streams to decrease average system latency by 68% in air-gapped environments.

Software Developer, Tecknotrove System Pvt Ltd, Mumbai, India

Jun 2022 - Nov 2023

- Implemented core C# application logic for enterprise training software, integrating a MySQL relational database to securely capture, store, and query thousands of user evaluation data points in real time.
 - Engineered a pixel-perfect scoring algorithm extracting UV texture coordinates via vector raycasting; replaced complex legacy physics colliders with 2D color-map sampling to cut CPU workload by 86% and eliminate clipping errors.
 - Executed a full-scale core engine migration (Unity 5.6 to 2019) for a legacy enterprise project, resolving 54 critical package dependencies and establishing strict hardware-to-software interfacing protocols.
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EDUCATION

Master of Science in Computer Science,

Aug 2025 - Expected Dec 2026

University at Buffalo, The State University of New York, Buffalo, NY

GPA: 3.75 / 4.0

Bachelor of Technology in Computer Science,

Jul 2018 - May 2022

Vellore Institute of Technology, India

CGPA: 7.72 / 10.0

PROJECTS

OpenStream | Java, Spring Boot, PostgreSQL, React, JUnit 5, Mockito, Docker | openstream.anutej.us

Sep 2026

- Built and deployed a Spring Boot REST API with 11 endpoints using a controller-service-repository architecture, Spring Data JPA, Hibernate, and PostgreSQL on Neon to manage users, posts, and follow relationships.
- Modeled a self-referential many-to-many follow graph with foreign key and composite-unique constraints; optimized chronological feed retrieval using JPQL JOIN FETCH to avoid N+1 queries.
- Developed 15 JUnit 5 and Mockito unit tests for validation, duplicate relationships, missing users, and service-layer business rules; containerized the API with a multi-stage Docker build and deployed it to Render.

BERTweet-Guard Moderation | Python, PyTorch, FastAPI, Hugging Face, NLP

Jan 2026 - May 2026

- Engineered a Natural Language Processing (NLP) text moderation model by fine-tuning a pretrained BERTweet foundation model via Low-Rank Adaptation (LoRA) on a 160,000-row Jigsaw dataset using PyTorch and Hugging Face.
- Formulated a weighted focal loss function to address class imbalance and calibrated the decision threshold via a validation macro-F1 sweep, achieving 0.917 Test macro-F1 and 97.1% test accuracy on the held-out evaluation set.
- Deployed the fine-tuned model as an asynchronous FastAPI inference service, merging LoRA adapter weights into the BERTweet backbone and loading production artifacts from Hugging Face for low-latency inference.

Interactive Portfolio | React, Vite, Spring Boot, Gemini API, GitHub Actions | anutej.us

Feb 2026

- Built a React/Vite SPA replicating the VS Code interface, including an in-memory virtual file system, responsive component rendering, dynamic tabs, and light/dark themes.
 - Developed a terminal emulator with a custom lexical parser supporting UNIX-style commands such as ls, cd, and open to navigate the virtual file system.
 - Built a Spring Boot REST proxy for the Gemini API with server-side credential management, per-IP rate limiting, and retry handling; automated frontend CI/CD to GitHub Pages with GitHub Actions.
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SKILLS

- **Languages:** Java, Python, C#, JavaScript, Go (Golang), C, SQL, HTML/CSS
 - **Backend & Data:** Spring Boot, Spring Web, Spring Data JPA, Hibernate, FastAPI, REST APIs, PostgreSQL, MySQL
 - **Frontend & DevOps:** React, Vite, Tailwind CSS, React Router, Docker, Git, GitHub Actions, Maven, Linux, Render
 - **Testing & AI/ML:** JUnit 5, Mockito, PyTorch, Hugging Face Transformers, LoRA, Pandas, NumPy, Ollama
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HONORS & AWARDS

National Winner, Smart India Hackathon 2020 | VR Fire Safety Simulator

Dec 2020